



May 9, 2018

Neuravi Ltd.
Mairsíl Claffey
VP Clinical, Quality and Regulatory Affairs
Block 3, Ballybrit Business Park
Galway, Ireland

Re: K173452

Trade/Device Name: EmboTrap® II Revascularization Device
Regulation Number: 21 CFR 870.1250
Regulation Name: Percutaneous Catheter
Regulatory Class: Class II
Product Code: NRY
Dated: April 6, 2018
Received: April 9, 2018

Dear Mairsíl Claffey:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820); and if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm> for the CDRH's Office of Surveillance and Biometrics/Division of Postmarket Surveillance.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/>) and CDRH Learn (<http://www.fda.gov/Training/CDRHLearn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<http://www.fda.gov/DICE>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Carlos L. Pena -S 

Carlos L. Peña, PhD, MS
Director
Division of Neurological
and Physical Medicine Devices
Office of Device Evaluation
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K173452

Device Name

EmboTrap® II Revascularization Device

Indications for Use (Describe)

The EmboTrap® II Revascularization Device is intended to restore blood flow in the neurovasculature by removing thrombus in patients experiencing ischemic stroke within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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510(k) Summary - K173452

I. SUBMITTER:

510(k) Owner: Neuravi Ltd.

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Date Prepared: May 09, 2018

II. DEVICE

Trade Name of Device: EmboTrap® II Revascularization Device

Common Name of Device: Catheter, Thrombus Retriever

Classification Name: 21 CFR 870.1250 – Class II

Product Code: NRY

III. PREDICATE DEVICE(S)

Trevo ProVue Retriever (K122478)

Solitaire™ FR Revascularization Device (K113455)

IV. DEVICE DESCRIPTION

The EmboTrap® II Revascularization Device is composed of a retrievable, self-expanding, Nitinol shaped section at the distal end of a tapered Nitinol shaft. It is designed to restore blood flow in the neurovasculature by removing thrombus in patients experiencing ischemic stroke. The EmboTrap® II Revascularization Device is supplied sterile, and is intended for single-use only by physicians trained in neuro-interventional catheterization and the treatment of ischemic stroke. It is indicated for patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy.

V. INDICATIONS FOR USE

The EmboTrap® II Revascularization Device is intended to restore blood flow in the neurovasculature by removing thrombus in patients experiencing ischemic stroke within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

VI. COMPARISON OF TECHNOLOGICAL CHARACTERISTICS WITH THE PREDICATE DEVICE

A summary of the technological characteristics of the EmboTrap® II Revascularization Device in comparison to those of the predicate device is presented below.

Characteristics	Predicate Devices Referenced in this Submission		Subject Device
	Trevo ProVue	Solitaire™ FR	EmboTrap® II Device
Manufacturer	Stryker	Covidien/Medtronic	Neuravi
510(k) Number	K122478	K113455	K173452
Classification	Class II (21CFR 870.1250)		Same
Device Classification Name	Catheter, Thrombus Retriever		Same
Classification Product Code	NRY		Same
Intended Use	Restoration of blood flow by removal of thrombus from the neurovasculature (large intracranial vessel) in patients experiencing ischemic stroke within 8 hours of symptom onset		Same
Target Population	Patients with symptoms of an ischemic stroke within 8 hours of symptom onset, who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment		Same
Design/Technological Principles	Self-expanding Nitinol shaped section Nitinol guide-wire like shaft		Same
Principal Device Materials			
Distal Shaped Section	Nitinol	Nitinol	Same
Core Wire (Shaft)	Nitinol	Nitinol	Same
Distal Marker/Coil	Platinum/Tungsten	Platinum/Iridium	Platinum/Tungsten
Proximal Marker/Coil	304 Stainless Steel	Platinum/Iridium	Platinum/Tungsten
Design Characteristics & Technology			
Size(s) Offered (Retriever Diameter × Length)	4×20 mm	4×15 mm, 4×20 mm, 6×20 mm, 6×30 mm	5×21 mm, 5×33 mm
Minimum Microcatheter ID	0.021”	0.021” (4 mm diameter size) 0.027” (6 mm diameter size)	0.021”
Additional Characteristics			
How Supplied	Sterile/Single Use	Sterile/Single Use	Same
Sterilization Method	Ethylene Oxide	Ethylene Oxide	Same

VII. PERFORMANCE DATA

Biocompatibility Testing:

The biocompatibility evaluation for the EmboTrap® II Revascularization Device was conducted in accordance with International Standard ISO 10993-1 "Biological Evaluation of Medical Devices – Part 1: Evaluation and Testing Within a Risk Management Process" as recognized by FDA (Recognition Number 2-156) and FDA Biocompatibility Guidance (Use of International Standard ISO 10993-1, "Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process", June 16, 2016).

Per ISO 10993-1, the EmboTrap® II device is categorized as an external communicating device with limited exposure, i.e. whose contact with circulating blood is less than 24 hours.

The biocompatibility evaluation included the following tests:

Test	Results	Conclusions
MTT Cytotoxicity Study	Based on the percentage viability values for the test article extract dilutions, the device is non-cytotoxic.	Device is non-cytotoxic per the MTT Cytotoxicity Study
ISO Guinea Pig Maximization Sensitization Test	Test article extracts showed no evidence of causing delayed dermal contact sensitization in the guinea pig.	Device is not considered a sensitizer per the Guinea Pig Maximization Test
ISO Intracutaneous Study in Rabbits	The difference between the test extract overall mean score and the corresponding control overall mean score was 1.0 or less.	Device is not an irritant when injected intracutaneously per the ISO Intracutaneous Study in Rabbits
ISO Systemic Toxicity Study in Mice	There was no mortality or evidence of systemic toxicity from the extracts injected into mice.	Device is non-toxic per the ISO Systemic Toxicity Study in Mice
USP Rabbit Pyrogen Study, Material Mediated	No individual rabbit showed a rise in temperature of $\geq 0.5^{\circ}\text{C}$ above its baseline temperature and the total maximum temperature rise of all three animals was within acceptable USP limits.	Device is non-pyrogenic per the Material Mediated Rabbit Pyrogen Study
ASTM Hemolysis	Both the test article in direct contact with blood and the test article extract were non-hemolytic.	Device is non-hemolytic per the ASTM Hemolysis Test
Complement Activation Assay Studies	The C3a and SC5b-9 concentrations of the test article samples were acceptable.	Levels of C3a and SC5b-9 were acceptable.
SC5b-9 Complement Activation Assay Study		

All biocompatibility tests completed met the pre-assigned acceptance criteria as specified in the test protocol and in accordance with the requirements of the applicable standards.

Sterilization and Shelf Life:

The EmboTrap® II Revascularization Device is labeled as a single-use, sterile device, with a shelf life of 3 years. The sterilization process for the EmboTrap® II device has been successfully validated and process monitoring controls are in place to assure that the device is EO-sterilized to achieve a minimum SAL of 10^{-6} .

Shelf life studies have been conducted for the EmboTrap® II device and establish that the product and packaging remain functional and sterile for the shelf life period of 3 years.

***In Vitro* (Bench) Testing:**

The bench testing performed to evaluate the EmboTrap® II device included the following tests:

- Dimensional Testing
- Radial Force Testing
- Outer Cage Recovery Testing
- Device Durability/Integrity Testing
- Joint Strength Testing
- Flexibility & Kink Resistance Testing
- Torque Durability Testing
- Corrosion Resistance Testing
- Radiopacity Testing
- Tip Flexibility Testing
- Re-sheathing Force Testing
- Simulated Use/Performance Testing
- Coating Integrity Testing
- Particulate Evaluation Testing

The results of design verification testing demonstrate that the device fulfils all pre-determined product performance specification requirements. In addition, the results of the side-by-side bench testing completed show that the EmboTrap® II device is comparable to its predicate devices in terms of performance and functionality. A summary of testing performed to evaluate key descriptive characteristics between the proposed and predicate devices is presented in the table below.

Characteristic/Test	Method	Conclusions
System Dimensions	Device attributes (overall device length, length and OD of the shaped section, and foreshortening) were measured compared with one or more predicate devices.	The subject device dimensions are within the range of existing predicate dimensions for this device type. The longer overall length of the subject device does not affect performance, safety or effectiveness.
Tip Flexibility	Tip flexibility was evaluated compared with a predicate device.	The subject device demonstrated greater flexibility than the predicate device tested at a 90° approach angle, and equivalent tip flexibility at a 70° approach angle.
Re-sheathing force	The force required to re-sheath the device was evaluated compared with one or more predicate devices.	The force required to re-sheath the subject device is within the same range as that measured for the predicate devices.
Delivery and re-sheathing force during simulated use	The delivery and re-sheathing forces measured during simulated use of the subject device were compared with the forces measured for one or more predicate devices.	The delivery and re-sheathing forces of the subject device are comparable to that of the predicate device tested during simulated use.
Durability Testing	Damage was evaluated after delivery and withdrawal of the device beyond the recommended number of passes and re-sheathings recommended in the instructions for use compared with one or more predicate devices.	Devices tested demonstrated no damage after delivery and withdrawal testing. Durability of the subject device is equivalent to that of the predicate device tested.
System Kink Resistance	Kink resistance of the entire device (shaft and shaped section) was evaluated when wrapped around a series of mandrels of decreasing radii until permanent deformation was observed or until the smallest radius was used.	Kink resistance of the shaped section was comparable to that of the predicate device tested.
Kink Resistance – Deployed Shaped Section	Kink resistance of the deployed shaped section was evaluated in a series of bend radii within a range of vessel lumen diameters.	Kink resistance of the deployed shaped section was equivalent to or superior than that of the predicate device tested.
Distal Coil (Markercoil) Tensile Testing	The tensile strength of the distal coil bond was evaluated and compared with one or more predicate devices.	The tensile strength of the distal marker coil met acceptance criteria and comparable to that of the predicate device.
System Tensile Testing	The peak tensile strength to failure of the fully assembled device was evaluated post-simulated use and compared with that measured for one or more predicate devices.	The system tensile strength of the subject device is equivalent to that of the predicate device tested.

Characteristic/Test	Method	Conclusions
Torque Testing in a Microcatheter	Devices were tracked through a microcatheter in a tortuous path anatomical model and evaluated for damage following a number of rotations with the distal end constrained.	Torque durability performance of the subject device is equivalent to that of the predicate device tested.
Torque Strength Testing	The number of rotations to separate the subject device when torqued in a representative tortuous model was compared with that measured for one or more predicate devices.	Torque durability performance of the subject device is better than that of the predicate device tested.
Radial Force Testing	The radial force of the subject device was measured within a range of lumen diameters applicable to the intended vasculature and compared with the radial forces measured for one or more predicate devices.	The radial force of the subject device, when tested in representative lumen sizes, is comparable to that of the predicate devices tested.
Radiopacity	Radiopacity of the subject device was evaluated fluoroscopically and compared with one or more predicate devices.	Radiopacity of the subject device is equivalent to that of the predicate devices tested.
Clot retrieval and performance	Performance of the subject device and its effectiveness at retrieval of firm and soft clots of various sizes in a tortuous anatomical model was evaluated and compared with one or more predicate device.	The subject device effectively retrieved clot and restored flow in the test model. Device performance was rated equivalent to or better than the predicate devices tested for loading, delivery, deployment and retrieval in an <i>in vitro</i> tortuous path anatomical model. <i>In vitro</i> clot retrieval performance of the subject device is comparable to or better than that of the predicate devices tested.
Physician usability testing	The user's ability to reliably deploy and use the subject device in a tortuous benchtop model was evaluated and compared with one or more predicate devices.	The subject device usability performance was rated higher than or equivalent to the predicate device tested for deliverability, deployment, retrieval, and re-sheathing.

***In Vivo* (Animal) Studies:**

Acute and chronic animal studies have been performed to assess the usability, effectiveness and safety of the EmboTrap® II Revascularization Device compared to the predicate devices in the swine model. Acute performance evaluated on Day 0 showed that the usability and performance of the EmboTrap® II device was equivalent to that of the predicate device tested. Histological evaluation performed on treated vessels after 3 and 28 days demonstrated that the local and end organ tissue response was comparable between the EmboTrap® II device and the predicate devices tested.

Clinical Studies:

Substantial equivalence of the EmboTrap® II Revascularization Device has been demonstrated through the results of the ARISE II (Analysis of Revascularization in Ischemic Stroke with EmboTrap) Study.

The ARISE II (Analysis of Revascularization in Ischemic Stroke with EmboTrap®) clinical trial assessed the efficacy and safety of the EmboTrap® Revascularization Device against a performance goal derived from the SWIFT and TREVO 2 clinical trials. Key inclusion criteria were: Patients with a large-vessel occlusion who could be treated within 8 hours of stroke symptom onset; age 18-85; $8 \leq \text{NIHSS} \leq 25$; Angiographic confirmation of an occlusion of an ICA, MCA, M1, or M2, vertebral or basilar arteries, and IV-tPA, if used, was initiated within 3 hrs of stroke onset. Key exclusion criteria were: Stenosis, or any occlusion, in a proximal vessel that required treatment or would prevent access to the site of occlusion.

Two hundred and forty-four (244) patients signed informed consent, but 16 were identified by the treating physician as not meeting the angiographic enrolment criteria. Of the 228 identified as suitable for treatment one became too agitated for treatment and was not treated with any device. Two hundred and twenty-seven (227) patients were treated with the EmboTrap® device and included in the study. Of these, 191 patients met all inclusion and exclusion criteria as requested by FDA for Intention-to-Treat (ITT) Cohort. FDA further requested that within this FDA ITT cohort, use of rescue therapy at any point in the procedure be imputed as a failure to achieve the revascularization and good clinical outcome endpoints. Among the 36 patients excluded to meet the FDA defined ITT criteria, some had more than one deviation from the criteria. Those removed include 12 Patients with IV-tPA initiated between 3-4.5hr, 5 missing pregnancy tests, 1 ≥ 86 yrs, 4 with treated carotid stenosis, 1 dissection, and 1 with evidence of multiple occlusions, as well as 12 who did not meet the criteria around laboratory ranges for GFR, Glucose, Platelets and/or INR.

ARISE II Primary Endpoints:

Primary Efficacy Endpoint

The primary effectiveness endpoint was revascularization measured using modified Thrombolysis in Cerebrovascular Infarction (mTICI inclusive of the 2c rating).

Successful achievement of the endpoint is defined as achieving an mTICI score of 2b or greater in the target vessel following 3 or less passes of the EmboTrap® device without the use of rescue therapy at any point during the procedure.

Primary Safety Endpoint

The primary safety endpoint was the occurrence of Symptomatic Intracerebral hemorrhage (sICH) within 24 hours (-8/+12 hrs) post-procedure, together with any other Serious Adverse Device Effects (excluding those already counted in sICH).

ARISE II Statistical Analysis:

The Primary Efficacy endpoint was designed to evaluate performance of the EmboTrap® device compared to predicate devices and was based on a one-sided Exact test for a binomial proportion of the hypothesis:

$H_0: P_{\text{EmboTrap}} \leq NL$ versus $H_1: P_{\text{EmboTrap}} > NL$, where $NL = 0.56$

ARISE II Study Procedures:

Angiographic data were obtained at baseline (prior to patient treatment), and again at the conclusion of each revascularization attempt, and after treatment. Twenty-four hour post-procedure follow-up included CT or MR imaging and NIHSS examination. Follow-up at Day 7 (± 12 hours) or discharge, if earlier, and at Day 90 (± 14 days) included NIHSS exam and modified Rankin Scale (mRS) evaluation. Adverse events were recorded throughout the hospitalization period and at each follow-up.

ARISE II Inclusion Criteria:

- The patient or the patient's legally authorized representative has signed and dated an Informed Consent Form.
- Aged between 18 years and 85 years (inclusive).
- A new focal disabling neurologic deficit consistent with acute cerebral ischemia.
- NIHSS score ≥ 8 and ≤ 25 .
- Pre-ictal mRS score of 0 or 1.
- The interventionalist estimates that at least one deployment of the EmboTrap® device can be completed within 8 hours from the onset of symptoms.
- Patients for whom IV-tPA is indicated and who are available for treatment, are treated with IV-tPA.
- IV-tPA, if used, was initiated within 3 hrs of stroke onset (onset time is defined as the last time when the patient was witnessed to be at baseline), with investigator verification that the subject has received/is receiving the correct IV t-PA dose for the estimated weight.
- Angiographic confirmation of an occlusion of an ICA (including T or L occlusions), M1 or M2 MCA, VA, or BA with mTICI flow of 0 – 1.
- For strokes in the anterior circulation, the following imaging criteria should also be met:
 - MRI criterion: volume of diffusion restriction visually assessed ≤ 50 mL; or
 - CT criterion: ASPECTS 6 to 10 on baseline CT or CTA-source images, or, volume of significantly lowered CBV ≤ 50 mL.
- The patient is indicated for neurothrombectomy treatment by the interventionalist and it is confirmed by diagnostic angiography that the device will be able to reach the target lesion proximally.

ARISE II Exclusion Criteria:

- Life expectancy likely less than 6 months.
- Females who are pregnant or breastfeeding.
- History of severe allergy to contrast medium.
- Known nickel allergy at time of treatment.
- Known current use of cocaine at time of treatment.
- Patient has suffered a stroke in the past 3 months.
- The patient presents with an NIHSS score <8 or >25 or is physician assessed as being in a clinically relevant uninterrupted coma.
- Subject participating in another study involving an investigational device or drug.
- Use of warfarin anticoagulation or any Novel Anticoagulant with International Normalized Ratio (INR) >3.0 .
- Platelet count $<50,000/\mu\text{L}$.
- Glucose <50 mg/dL.
- Any known hemorrhagic or coagulation deficiency.
- Unstable renal failure with serum creatinine >3.0 or Glomerular Filtration Rate (GFR) <30 .
- Patients who have received a direct thrombin inhibitor within the last 48 hours; must have a partial thromboplastin time (PTT) less than 1.5 times the normal to be eligible.
- All patients with severe hypertension on presentation (SBP >220 mmHg and/or DBP >120 mm Hg). All patients, in whom intravenous therapy with blood pressure medications is indicated, with hypertension that remains severe and sustained despite intravenous antihypertensive therapy (SBP >185 mmHg and/ or DBP >110 mmHg).
- Known cerebral vasculitis.
- Rapidly improving neurological status.
- Clinical symptoms suggestive of bilateral stroke or stroke in multiple territories.
- Ongoing seizure due to stroke.
- Evidence of active systemic infection.
- Known cancer with metastases.
- Computed tomography (CT) or Magnetic Resonance Imaging (MRI) evidence of recent/ fresh hemorrhage on presentation.
- Baseline computed tomography (CT) or MRI showing mass effect or intracranial tumor (except small meningioma).
- Suspicion of aortic dissection, presumed septic embolus, or suspicion of bacterial endocarditis.
- Stenosis, or any occlusion, in a proximal vessel that requires treatment or prevents access to the site of occlusion.
- Evidence of dissection in the extra or intracranial cerebral arteries.
- Occlusions in multiple vascular territories (e.g., bilateral anterior circulation, or anterior/posterior circulation).

ARISE II Results:

The ARISE II (Analysis of Revascularization in Ischemic Stroke with EmboTrap®) clinical trial assessed the efficacy and safety of the EmboTrap® Revascularization Device against a performance goal derived from the SWIFT and TREVO 2 clinical trials. Patients with a large-vessel occlusion who could be treated within 8 hours of stroke symptom onset were eligible for enrolment. Of the 227 patients treated with the EmboTrap® device, 191 patients met all inclusion and exclusion criteria (FDA Intention-to-Treat (ITT) Cohort).

The Primary Efficacy Endpoint was successful revascularization defined as an mTICI score of at least 2b in the target vessel, following three or less passes of the EmboTrap® device without rescue, using Core Laboratory adjudicated data. For the FDA ITT cohort, successful revascularization was modified to count any case in whom rescue therapy was used, after the EmboTrap® device, as a failure to meet the primary efficacy endpoint.

The Primary Safety Endpoint was the occurrence of symptomatic intracerebral hemorrhage (sICH) within 24 hours of the procedure, together with any serious adverse device effects (SADE), excluding those already counted as sICH. The Primary Safety Endpoint was adjudicated by the Clinical Events Committee (CEC).

ARISE II Study Results	FDA ITT Cohort (N=191)
Primary Efficacy Endpoint	
Successful Revascularization¹ [95% Conf. Interval] ² p-value	72% (137 ^b /191) [65%, 78%] <0.0001 ^a
Primary Safety Endpoint	
Occurrence of sICH within 24h and/or SADE [Unadjusted 95% Conf. Interval*]	5% (9 ^c /191) [2%, 9%]
Key Secondary Endpoints	
Good clinical outcome³ [Unadjusted 95% Conf. Interval*]	53% (102 ^{b,d} /191) [46%, 61%]
Procedure-related mortality at 7 days post-procedure [Unadjusted 95% Conf. Interval*]	0% (0/191) [0%, 2%]
All-cause mortality at 90 days post-procedure [Unadjusted 95% Conf. Interval*]	11% (21 ^d /191) [7%, 16%]
Neurological deterioration⁴ [Unadjusted 95% Conf. Interval*]	9% (18 ^d /191) [6%, 14%]
Procedure Related Serious Adverse Events (PRSAE) [Unadjusted 95% Conf. Interval*]	5% (9/191) [2%, 9%]

1. Successful revascularization was defined as achieving an mTICI rating of 2b or greater, in 3 or less passes of the study device.

2. All confidence intervals reported for binary variables are Clopper-Pearson Exact binomial confidence intervals, and those for continuous variables are based on a normal distribution assumption for the parameter of interest; 3. Modified Rankin Scale score of ≤2 at 90 days post-procedure (missing patients imputed as failure); 4. Increase of ≥4 points on the NIHSS score at 24 hours post-procedure.

**The confidence intervals are calculated without multiplicity adjustment. As such, the confidence intervals are provided to show the variability only and should not be used to draw any statistical inferences.*

a. One-sided Exact test at the two-sided 5% significance level for the revascularization rate against a performance goal of 56%;

b. Use of rescue therapy at any point in the procedure was imputed as a failure to achieve the endpoint; c. No occurrence of SADE; d. Unknown/Missing data were imputed as worst-case (this affected 4 subjects for Good Clinical Outcome, 8 subjects for Neurological Deterioration, and 5 subjects for All-Cause Mortality at 90 Days).

Primary Efficacy Endpoint by Region:

Population: FDA ITT Cohort (N=191) Primary Efficacy Endpoint	US cohort (N=94)	EU cohort (N=97)
Successful Revascularization ^{1,2} [95% Conf. Interval] ³	67% (63/94) [57%, 76%]	76% (74/97) [67%, 84%]

1. Successful revascularization was defined as achieving an mTICI rating of 2b or greater, in 3 or less passes of the study device.

2. Use of rescue therapy at any point in the procedure was imputed as a failure to achieve the endpoint; 3. All confidence intervals reported for binary variables are Clopper-Pearson Exact binomial confidence intervals, and those for continuous variables are based on a normal distribution assumption for the parameter of interest.

ARISE II Summary of Serious Adverse Events

One hundred and seventeen (117) SAE occurred within the FDA ITT Cohort. The most common (incidence of $\geq 1\%$) SAE are presented below.

Serious Adverse Events occurring at an incidence of 1% or more in any population	
MedDRA Preferred Term	% pts w/ SAE [Number of SAE]
	FDA ITT Cohort (N=191)
Cardiac disorders	
Myocardial infarction	1.57% [3]
Atrial fibrillation	1.05% [2]
Cardiac failure	1.05% [2]
Gastrointestinal disorders	
Gastrointestinal haemorrhage	1.05% [2]
Infections and infestations	
Septic shock	2.09% [4]
Urinary tract infection	1.57% [3]
Pneumonia	1.57% [3]
Sepsis	1.05% [2]
Nervous system disorders	
Ischaemic stroke	6.81% [13]
Haemorrhagic transformation stroke	6.28% [12]
Brain oedema	3.66% [7]
Neurological decompensation	2.09% [4]
Seizure	1.57% [3]
Subarachnoid haemorrhage	1.57% [3]
Carotid artery stenosis	1.05% [2]
Respiratory, thoracic and mediastinal disorders	
Pneumonia aspiration	1.57% [3]
Respiratory distress	1.05% [2]
Bronchospasm	1.05% [2]
Respiratory failure	1.05% [2]
Surgical and medical procedures	
Coronary arterial stent insertion	1.05% [2]
Vascular disorders	
Vessel perforation	1.05% [2]
Deep vein thrombosis	1.05% [2]

Reasons for exclusion from treatment:

Two-hundred and forty-four (244) patients were consented for enrolment in the ARISE II Study, of which, 17 were excluded from treatment for the reasons outlined below.

Angiographic reasons for exclusion from treatment	Number of Subjects (N=16)*
No angiographic confirmation of an occlusion of an ICA (including T or L occlusions), M1 or M2 MCA, VA, or BA with mTICI flow of 0 – 1.	7
The patient is indicated for neurothrombectomy treatment by the interventionalist and it is confirmed by diagnostic angiography that the device will not be able to reach the target lesion proximally.	7
Stenosis, or any occlusion, in a proximal vessel that requires treatment or prevents access to the site of occlusion.	4
Evidence of dissection in the extra or intracranial cerebral arteries.	2
Occlusions in multiple vascular territories (e.g., bilateral anterior circulation, or anterior/posterior circulation).	2
Other reasons for exclusion from treatment	Number of Subjects (N=1)
Severe agitation	1

***7 patients failed 2 angiographic criteria.**

In addition to the pre-specified performance goal evaluation of the ARISE II study, FDA also considered the ARISE II revascularization rates (mTICI scores) relative to more recent studies of stent retriever devices in acute ischemic stroke patients as additional supportive information in the substantial equivalence assessment.

CONCLUSIONS

Non-clinical and clinical studies demonstrate that the EmboTrap® II Revascularization Device is substantially equivalent to the predicate devices.